

Open-Set Recognition for Amazon Tree Crown Species Identification from UAV Imagery

1. Introduction

Accurate identification of tree species from unmanned aerial vehicle (UAV) imagery is a critical component of precision forestry management in the Amazon. Traditional image classification approaches operate under a **closed-set assumption**: the model is expected to assign every input to one of a fixed set of known classes encountered during training. In real forest environments, however, a deployed model will inevitably encounter species that were not represented in the training data — whether due to incomplete annotation, regional biodiversity, or seasonal variation in forest composition.

Open-Set Recognition (OSR) addresses this limitation by equipping a model to simultaneously classify known species correctly and *reject* samples that do not belong to any known class, flagging them as unknown. This capability is particularly important in precision forest inventory, where silently misclassifying an unknown species as a known one could distort timber volume estimates and conservation assessments.

Research Scope and Hypotheses

This study investigates OSR for Amazon tree crown identification using UAV-derived orthophotos. We make the following assumptions explicit:

1. **Pseudo-label setting.** Ground-truth species labels for individual tree crowns at scale are prohibitively expensive to obtain. We use predictions from NetFlora — an open-source, Embrapa-developed drone-based forest species detector — as proxy labels. This means our conclusions concern *relative improvements* between methods under a consistent labelling scheme, not absolute species identification accuracy.
2. **Experimental known/unknown split.** The division of species into "known" (seen during training) and "unknown" (held out entirely) is an experimental construct. We selected the five most abundant, well-balanced species as known classes and six additional species as unknown classes based on detection statistics (Section 2).

Contributions

1. A dual-year, pseudo-labelled tree crown OSR benchmark with 5 known and 6 unknown Amazon species classes.
 2. A systematic comparison of 4 post-hoc OSR scoring methods, demonstrating a clear advantage of feature-space methods over logit-based methods.
 3. A Supervised Contrastive Learning (SupCon) extension that improves OSR performance without requiring additional data.
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2. Dataset Construction and Experimental Setup

2.1 Data Sources

Imagery. We use UAV orthophotos acquired over managed forest concessions in Acre state, Brazil, during two campaigns:

- **2023:** 179 TIF tiles across 22 flight areas (2023-01 to 2023-22)
- **2025:** 55 TIF tiles across 2 flight areas (2025-11, 2025-12)

All tiles are RGBA, approximately 3.4–3.9 cm GSD, stored in UTM coordinates (EPSG:32718 for 2023, EPSG:32719 for 2025).

NetFlora. NetFlora (Embrapa Acre) is a YOLOv7-based detector trained on Amazon drone imagery to recognise 23 tree species of commercial value. We use the public weights `PMFS_Emrapa00.pt`. NetFlora detections serve as our sole source of species labels; no manual annotation is performed beyond the NetFlora training data already incorporated into the model.

2.2 Dataset Construction Pipeline

The full construction pipeline proceeds in 3 steps:

Step 1 — Full scan. All TIFs are scanned exhaustively with a non-overlapping 1024×1024 sliding window (stride = 1024, margin 5%). NetFlora is run at $\text{conf} \geq 0.45$ (known) or ≥ 0.55 (unknown), retaining only the 11 selected classes. Total raw detections: **6,801** (known) + **1,837** (unknown, after $\text{conf} \geq 0.55$ filter).

Class	2023	2025	Mean conf 2023	Mean conf 2025
CUMARU	1509	897	0.41	0.47
COPAIBA	544	513	0.40	0.45
GARAPA	468	489	0.44	0.44
MANITE	313	775	0.44	0.47
PINHO	360	933	0.47	0.49
CEDRO	455	225	0.47	0.56
CASTANHEIRA	178	262	0.40	0.47
CAUCHO	327	383	0.46	0.58
TAXI	209	161	0.49	0.54
MACARANDUBA	115	199	0.41	0.43
TAUARI	99	273	0.37	0.41
(others)	...	...		

Selection criteria for known classes: species where both years yield ≥ 300 detections at $\text{conf} > 0.45$, ensuring sufficient training samples after balancing. This yields five known classes: **COPAIBA, CUMARU, GARAPA, MANITE, PINHO**.

Selection criteria for unknown classes: six non-core species with $\text{conf} > 0.55$ and ≥ 50 detections in both years: **CAUCHO, CASTANHEIRA, CEDRO, MACARANDUBA, TAUARI, TAXI**. The higher confidence threshold (0.55 vs 0.45) is applied to unknowns to minimise pseudo-label noise in the held-out evaluation set.

Step 2 — TIF-level train/test split. To prevent tile-level leakage — where crops from the same tree crown could appear in both train and test — we split at the **TIF level**. For each year independently, 20% of TIFs are held out as the test partition. This guarantees that every crown in the test set comes from a geographic region unseen during training.

Step 4 — Soft-balanced crop extraction. Within each (split, class, year) cell, detections are down-sampled to a maximum of **300 per cell** for training (no cap for test). Each retained detection is cropped from the original TIF as a square region around the NetFlora bounding box with 20% padding, resized to 224×224 RGB.

2.4 Final Dataset Statistics

Known classes (train + test):

Split	Class	2023	2025
train	COPAIBA	94	300
train	CUMARU	117	300
train	GARAPA	220	300
train	MANITE	158	300
train	PINHO	166	300
test	COPAIBA	52	177
test	CUMARU	154	298
test	GARAPA	74	103
test	MANITE	66	212
test	PINHO	37	256

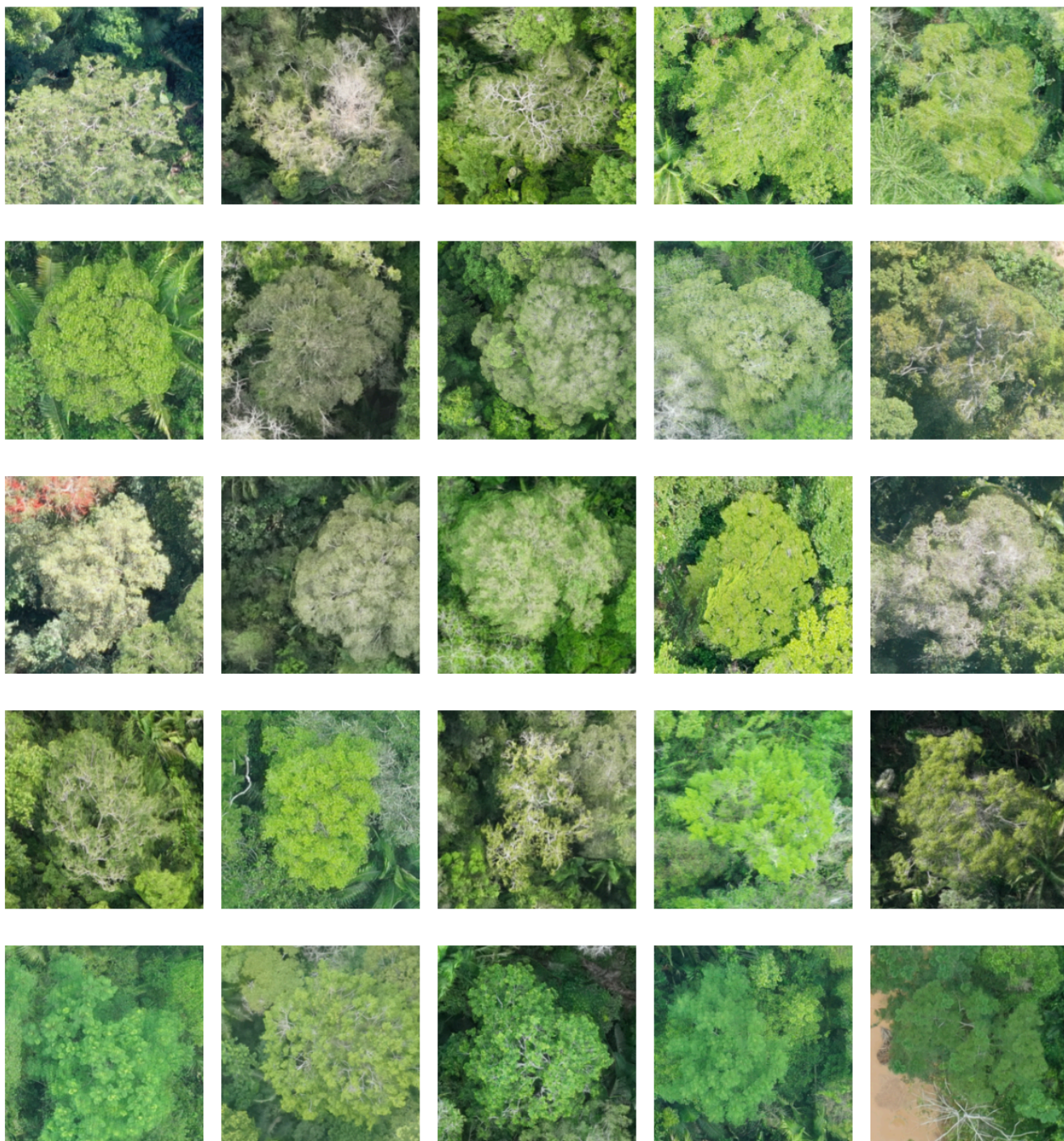
Unknown classes (test only, $\text{conf} \geq 0.55$):

Class	2023	2025
CAUCHO	98	200
CEDRO	99	172
CASTANHEIRA	34	163

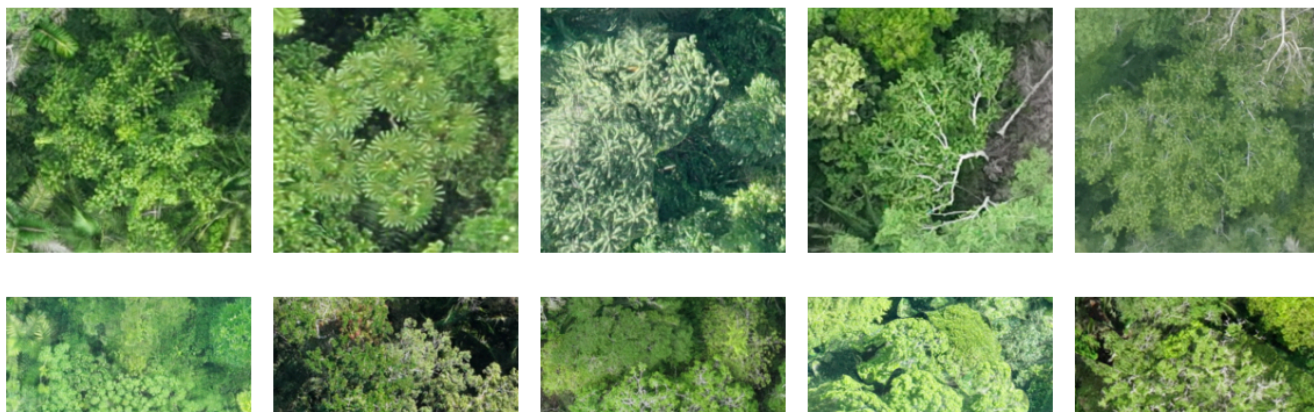
Class	2023	2025
TAXI	92	103
MACARANDUBA	53	137
TAUARI	38	151

CrownShift Dataset — Known and Unknown Tree Crown Examples

KNOWN CLASSES



UNKNOWN CLASSES





■ Known (train + test)
 ■ Unknown (test only, conf ≥ 0.55)

3. Methods

This section describes the full methodology: the backbone architecture, the two-stage closed-set classifier (baseline), the Supervised Contrastive Learning extension (SupCon), and the eight post-hoc OSR scoring methods applied to the trained model.

3.1 Backbone: DINOv2 ViT-B/14

We use **DINOv2 ViT-B/14** as the feature extractor throughout all experiments. DINOv2 is a Vision Transformer (ViT) pre-trained via self-supervised learning on a curated dataset of 142 million images, producing semantically rich, spatially aware representations without any task-specific supervision.

The ViT-B/14 variant processes images as non-overlapping 14×14 pixel patches. For a 224×224 input, this yields 256 patch tokens plus one [CLS] token. We use the [CLS] token output as the image representation, giving a **768-dimensional feature vector** per image.

All inputs are resized to 224×224, converted to RGB, and normalised with ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

3.2 Closed-Set Classifier (Baseline)

The baseline model is a **DINOv2 feature extractor with a lightweight MLP classification head**, trained to distinguish the five known species.

Architecture

```
Input (224×224 RGB)
  ↓
DINOv2 ViT-B/14 backbone      → 768-d [CLS] feature
  ↓
MLP head:
  Linear(768 → 256)
  ReLU
  Dropout(0.3)
  Linear(256 → 5)              → 5-class logits
```

The MLP head is intentionally lightweight (two layers) to avoid overfitting on top of the strong pre-trained features.

Two-Stage Training

Training proceeds in two stages to balance stability and adaptation:

Stage 1 — Frozen backbone, head training only

The DINOv2 backbone is fully frozen. Only the MLP head parameters are updated. This allows the head to adapt to the five-class task without disturbing the pre-trained representations.

Stage 2 — Partial backbone fine-tuning

The last 4 transformer blocks of the DINOv2 backbone are unfrozen and fine-tuned together with the head, using a much smaller learning rate to preserve pre-trained knowledge while adapting to the target domain.

The motivation for Stage 2 is empirical: fine-grained tree crown classification from UAV imagery requires domain-specific low-level feature adaptation. In our experiments, Stage 2 alone improved closed-set accuracy from 0.52 to 0.75.

Training Data

A **weighted random sampler** is used during training to compensate for any remaining class imbalance. A **validation split** (15% of training data, stratified by class and year) is held out for early stopping; it is never used for model selection across methods.

Training data augmentation includes random horizontal/vertical flips, rotation ($\pm 20^\circ$), and mild colour jitter. Test-time transforms are resize and normalise only.

3.3 Supervised Contrastive Learning Extension (SupCon)

Motivation

The baseline classifier is trained solely with CrossEntropy loss, which optimises for correct class assignment among the five known classes. It places no explicit constraint on the *geometry* of the feature space: two samples of different known classes could occupy nearby regions of feature space, as long as the linear head can separate them.

Supervised Contrastive Learning directly addresses this by adding an auxiliary loss that:

- **Pulls together** features from samples of the same known class
- **Pushes apart** features from samples of different known classes

The result is a feature space where known-class clusters are compact and well-separated, leaving larger gaps between and around clusters. Unknown species, having no cluster of their own, naturally fall into these gaps — making them easier to detect via KNN or Mahalanobis distance.

Projection Head

SupCon operates on a **128-dimensional unit sphere** rather than directly on the 768-d DINOv2 features. A small projection head maps features to this space:

```
DINOv2 [CLS] feature (768-d)
  ↓
Linear(768 → 512)
ReLU
Linear(512 → 128)
L2 normalise                → 128-d unit-sphere projection z
```

This follows standard contrastive learning practice: the projection head learns a representation optimised for the contrastive objective, while the backbone and original 768-d features remain available for downstream tasks. Crucially, **all OSR scoring at test time uses the original 768-d DINOv2 features**, not the projection. The projection head is discarded after training.

SupCon Loss

For a mini-batch of N samples with features $\{z_i\}$ and labels $\{y_i\}$, the SupCon loss is:

$$\mathcal{L}_{\text{SupCon}} = \sum_{i=1}^N \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \neq i} \exp(z_i \cdot z_a / \tau)}$$

where $P(i) = \{j \neq i : y_j = y_i\}$ is the set of positive pairs (same class as i), and $\tau = 0.1$ is the temperature. The loss averages over all positive pairs in the batch, encouraging each sample to be closer to all same-class samples than to any different-class sample.

Combined Training Objective

SupCon is introduced in Stage 2, added to the CrossEntropy loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{SupCon}}$$

where $\lambda = 0.5$ balances the two objectives. Stage 1 remains CrossEntropy-only (head warmup), ensuring stable initialisation before the contrastive objective is introduced.

The projection head parameters are trained jointly with the backbone and classification head in Stage 2, then discarded.

3.4 Post-hoc OSR Scoring Methods

Given a trained model (either baseline CE or SupCon), OSR is performed **without any retraining** by computing a scalar *unknownness score* for each test sample. A higher score means the sample is more likely to be a known class; a lower score means it is more likely unknown. We compare eight methods spanning two families.

Family 1: Logit-based methods

These methods operate on the 5-dimensional logit vector $\ell \in \mathbb{R}^5$ output by the classification head. Let $p = \text{softmax}(\ell)$.

MSP (Max Softmax Probability)

$$s_{\text{MSP}} = \max_c p_c$$

The simplest baseline. Known samples are expected to have high confidence in one class; unknown samples should be uncertain. In practice, deep networks are over-confident on unknown inputs, making MSP a weak OSR detector.

MaxLogit

$$s_{\text{MaxLogit}} = \max_c \ell_c$$

Operates on raw logits rather than softmax probabilities, preserving magnitude information that softmax normalisation discards.

Family 2: Feature-based methods

These methods operate on the 768-dimensional DINOv2 feature vector $f \in \mathbb{R}^{768}$, extracted before the classification head. They require access to the training set features at inference time.

Mahalanobis (Centroid) Distance

For each known class c , compute the class centroid μ_c from all training features of that class. Fit a single shared precision matrix Σ^{-1} from the pooled within-class scatter. The score for a test feature f is:

$$s_{\text{Maha}}(f) = -\min_c (f - \mu_c)^\top \Sigma^{-1} (f - \mu_c)$$

The negative minimum Mahalanobis distance to any known class centroid. Known samples are expected to be close to their class centroid; unknown samples are far from all centroids.

KNN (k-Nearest Neighbour Distance)

Store all N_{train} training features after L2-normalisation. For a test feature $\hat{f}$ (also L2-normalised), find its k nearest neighbours in the training set and compute:

$$s_{\text{KNN}}(f) = -\frac{1}{k} \sum_{i=1}^k \|\hat{f} - \hat{f}_{\text{train}}^{(i)}\|_2, \quad k = 5$$

The negative mean distance to the k nearest training neighbours. Unlike Mahalanobis, KNN makes no parametric assumption about the shape of class distributions, making it more robust when clusters are non-Gaussian or multi-modal. Known samples fall within the dense training distribution; unknown samples occupy sparse regions.

Summary Table

Method	Input	Parametric	Key assumption
MSP	logits	No	Unknown → low max softmax
MaxLogit	logits	No	Unknown → low max logit
Mahalanobis	features	Yes	Unknown → far from class centroids
KNN	features	No	Unknown → sparse training neighbourhood

4. Evaluation Metrics

We evaluate each method across three dimensions: closed-set classification quality, open-set detection ability, and (as a supporting analysis) year-domain separability. Each dimension requires different metrics because the tasks have different structures.

4.1 Closed-Set Metrics

Accuracy

$$\text{Acc} = \frac{\text{number of correctly classified known-class samples}}{\text{total known-class test samples}}$$

Straightforward fraction of known-class test crops assigned to the correct species. Reported as a sanity check; macro-F1 is the primary closed-set metric.

Macro-averaged F1 (macro-F1)

For each class c , compute the F1 score:

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

Then average equally across all five known classes:

$$\text{macro-F1} = \frac{1}{5} \sum_{c=1}^5 F1_c$$

Macro-F1 is preferred over accuracy because it weights each class equally regardless of its test-set size.

4.2 Open-Set Metrics

Both metrics below treat the OSR problem as a ranking/threshold problem: given a scalar score $s(x)$ for each test sample (higher = more known-like), how well can we separate known from unknown samples?

AUC (Area Under the ROC Curve)

The ROC curve plots True Positive Rate (TPR, fraction of known samples correctly accepted) against False Positive Rate (FPR, fraction of unknown samples incorrectly accepted) as the acceptance threshold varies. AUROC is the area under this curve:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR})$$

AUC has a convenient probabilistic interpretation: it equals the probability that a randomly chosen known-class sample receives a higher score than a randomly chosen unknown sample. Key reference points:

AUC = 0.5 → no better than random (score is uninformative)

AUC = 1.0 → perfect separation (all known > all unknown)

AUC = 0.8 → a known sample scores higher than an unknown 80% of the time

AUC is **threshold-free**: it measures discriminability across all possible operating points, making it comparable across methods without selecting a specific rejection threshold. It is the primary OSR metric in this study.

5. Experimental Results

5.1 Closed-Set Classifier: Effect of Two-Stage Training

The first experiment validates the necessity of Stage 2 backbone fine-tuning. Both stages use the same mixed training set (2023 + 2025, both years combined).

Training	Closed-set Acc	Macro-F1
Stage 1 only (frozen backbone)	0.520	0.515
Stage 1 + Stage 2 (fine-tuned)	0.748	0.741

Stage 2 fine-tuning (unfreezing the last 4 transformer blocks, LR = 1e-5) improves closed-set accuracy by 0.23 absolute. This confirms that fine-grained tree crown classification from UAV imagery requires domain-specific adaptation of the DINOv2 backbone; a frozen feature extractor with a linear head is insufficient.

All subsequent experiments use the two-stage training protocol.

5.2 Post-hoc OSR Methods: Systematic Comparison

Using the two-stage classifier trained on both years (2023 + 2025), we evaluate all post-hoc OSR scoring methods.

Closed-set accuracy of the model: **0.748**.

Method	Family	AUC	OSCR
KNN	Feature	0.801	0.605
Mahalanobis	Feature	0.785	0.590
MaxLogit	Logit	0.682	0.552
MSP	Logit	0.606	0.501

5.3 Supervised Contrastive Learning: Effect on OSR

We compare the baseline CE-only classifier against the SupCon-augmented model, both trained on 2023 + 2025. OSR is evaluated with Mahalanobis.

Results are reported separately for 2023 and 2025 test years to reveal any year-consistency effects.

Model	Closed-set Acc	Maha AUC
Baseline (CE)	0.748	0.785
SupCon	0.780	0.804

6. Open-Set Ceiling Analysis: Per-Class Unknown Difficulty

6.1 Per-Class AUROC

The overall OSR AUROC plateaus at approximately 0.80 across post-hoc methods and training objectives. To understand why, we decompose the unknown test set by species and compute a per-class AUROC: each unknown species is evaluated individually against the full known-class test set using Mahalanobis scores from the SupCon model.

Unknown class	n	Per-class AUROC	Nearest known (confused with)	Fraction
CEDRO	372	0.917	PINHO	42%
CAUCHO	387	0.916	PINHO	32%
TAXI	236	0.898	PINHO	50%
TAUARI	189	0.698	GARAPA	51%
MACARANDUBA	198	0.618	MANITE	47%

Unknown class	n	Per-class AUROC	Nearest known (confused with)	Fraction
CASTANHEIRA	261	0.608	MANITE	41%

The six unknown classes split cleanly into two groups:

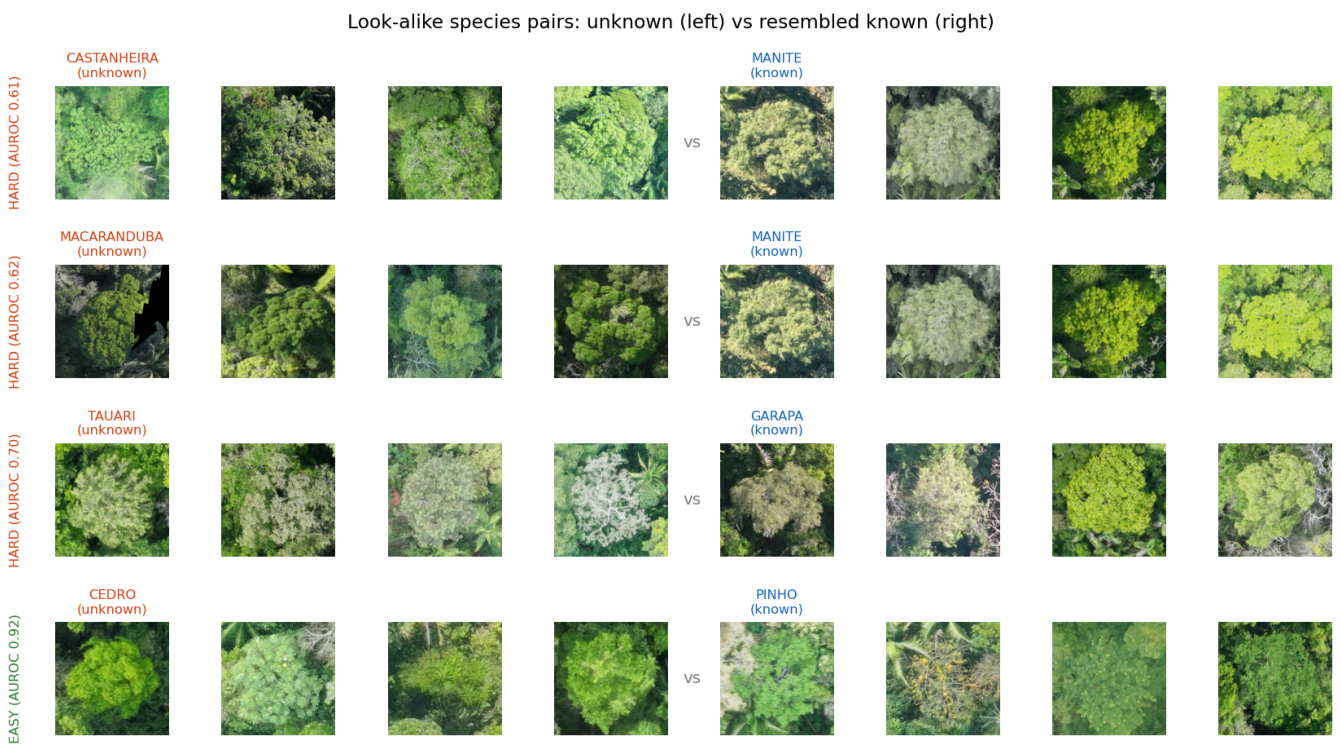
Easy unknowns (AUROC > 0.85): CEDRO (0.917), CAUCHO (0.916), TAXI (0.898). These three classes are highly detectable; their Mahalanobis scores fall well below the known-class score distribution. If OSR were evaluated on these three classes alone, the overall AUROC would exceed 0.90.

Hard unknowns (AUROC < 0.70): CASTANHEIRA (0.608), MACARANDUBA (0.618), TAUARI (0.698). These three classes are responsible for the 0.80 ceiling. Their Mahalanobis scores overlap substantially with the known-class distribution — the model cannot reliably distinguish them from known species.

6.2 Look-alike Species Pairs

The "confused with" column identifies which known class each hard unknown most resembles in feature space (nearest known centroid). The three hard pairs are:

TAUARI	(unknown)	↔	GARAPA	(known)	51% of TAUARI nearest to GARAPA
CASTANHEIRA	(unknown)	↔	MANITE	(known)	41% of CASTANHEIRA nearest to MANITE
MACARANDUBA	(unknown)	↔	MANITE	(known)	47% of MACARANDUBA nearest to MANITE



7. Rare Species Exp: (Next)

Netflora rare detection results:

Species	2023	2025
JUTAI	28	22
LOURO	5	43
ORELHA	7	4
BAJAO	23	36
ANGELIN	32	120
SAMAUMA	41	72